Project 4: Help Desk Ticket Analysis

What this project does

This project uses Python to analyze help desk tickets and create charts about ticket volume, type, priority, status, and resolution time.

Files used

The analysis script reads these files from the `Help Desk Tickets` folder:

- `issues.csv`
- `issues_change_history.csv`
- `issues_snapshot.csv`

How to run the analysis

1. Open the `Help Desk Tickets` folder in PyCharm.
2. Open `Project 4.py`.
3. If PyCharm says a package is missing, install `pandas` and `matplotlib` in the project's Python environment.
4. Run `Project 4.py`.

The script reads the data from the Desktop `Help Desk Tickets` folder. It saves chart images to PyCharm's working folder. To save new charts in `Help Desk Tickets`, set PyCharm's Working directory to that folder in the run settings.

The folder currently contains 14 analysis charts. The project checklist mentions 13, so check which count your assignment expects.

Dataset Source and Attribution

The help desk ticket data is from:

Abdellatif, Mohammad. *Help Desk Tickets*. Mendeley Data, Version 3. DOI: 10.17632/btm76zndnt.3.

Dataset page: https://data.mendeley.com/datasets/btm76zndnt/3

The dataset contains masked information to protect the data provider's privacy. The Mendeley Data record lists the dataset under the Creative Commons Attribution 4.0 International (CC BY 4.0) license.

If your download record shows a different version, use the DOI for that version instead.

Help Desk Ticket Analysis: Key Findings

1. Many tickets have no known priority

Of the 43,352 tickets reviewed, 25,989—about 60%—are marked "Unknown" for priority. This makes it harder to understand how urgent the tickets were and compare work across priority levels.

Recommendation: Check why so many tickets have an unknown priority. Consider requiring a priority when new tickets are created, where that makes sense for the help desk.

2. Resolution times vary widely

For the 43,048 tickets with resolution dates, the average time to resolve was 864.04 days, while the median was 45.46 days. The large difference suggests that a smaller number of tickets had very long calendar resolution times. Calendar time does not show how much active work was spent on a ticket.

Recommendation: Review the longest-running tickets and check whether their dates represent real resolution times. The data alone does not explain why they took so long.

3. Known priority levels are associated with different resolution times

Among tickets with a known priority, the median resolution time ranged from 3.75 days for Highest priority to 9.05 days for Low priority. This shows an association between assigned priority and resolution time; it does not prove that priority caused the difference or that the process was effective.

Recommendation: Review this pattern alongside workload, ticket type, and service targets before making decisions about team performance.

4. Unknown-priority tickets need further investigation

Unknown-priority tickets had a median resolution time of 1,400.40 days, far longer than tickets with known priorities. This may reflect older records, data migration, missing priority information, or other differences. The analysis does not establish the cause.

Recommendation: Compare ticket creation and resolution dates, priority-field history, closure patterns, and ticket types. This can help determine whether the long durations reflect actual service delays or historical/data-recording issues.

Overall conclusion

The data shows a substantial gap in priority information and a highly uneven spread of resolution times. More investigation is needed before treating long durations as evidence of current service delays or using the results to judge operational performance.